How Reproducible Are AI-Memory Benchmark Scores?
A Controlled Judge-and-Protocol Sensitivity Surface for LoCoMo

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DRAFT v0.1 (2026-05-31). Paper 2 — measurement sibling of the LoCoMo Audit (Paper 1). NOT FOR SUBMISSION without Stage-2 (paid) results.

Stage-1 cost: **$0.00**, **0** LLM/Anthropic calls (manifest: `total_cost_usd=0.0, llm_calls=0, zero_cost_asserted=true`).

Abstract

Commercial memory systems for large language models report wildly divergent accuracies on LoCoMo — the de-facto benchmark for very-long-term conversational memory — and the companion LoCoMo Audit [Gurbanov, 2026] shows *forensically* that the famous 84/58.44/75.14/92.5 spread decomposes into four methodological forks rather than four architecture verdicts. That audit's central empirical claim — *"small protocol changes swing the reported number"* — has until now been **asserted** but never **measured** as a controlled response surface. This paper turns the assertion into a measurement. We treat the evaluation protocol as a **factorial design** over six axes that are *not part of the system under test* — judge model, judge prompt, answer prompt, adversarial-category/denominator handling, retrieval-k, and seed — and quantify, per axis, how far a **fixed** system's reported score moves (the *sensitivity surface*), plus whether the *system ranking* itself flips under judge-only changes (rank stability, Kendall's τ).

Stage 1 (free, $0, zero LLM calls) measures the three axes computable by re-aggregating existing scored outputs on real LoCoMo: the largest single swing is the **retrieval-k axis** — holding the system fixed and moving k from 1 to 10, answer-evidence recall swings **+0.3221 (21.7% → 53.9%, a 32.2-point ∼2.5× move)** for tf-idf and **+0.3039 (25.6% → 56.0%)** for bm25; the **metric axis** flips a fixed system between **0.0% (exact-match) and 7.3% (normalized-contains)** on *identical* predictions; and the **Cat-5 denominator fork** is a deterministic **−22.5% relative haircut** (multiplier 1540/1986 = 0.7754, the **denominator component** of Paper 1's Fork A: ×0.7754 alone takes 84 → 65.1, and the residual to 58.44 is an **unmeasured** Stage-2 judge effect, not this fork). **None** of these touch the system under test. **Stage 2 (paid, $0.92 actual, pilot scope)** adds the judge axis: judge-prompt swing 27.5 pts on longcontext; ranking stable (τ=1.0) in this wide-gap pilot. Answer-prompt and seed axes remain **PENDING**.

1 Introduction *[DRAFTED]*

A reader who asks "which LLM memory system is best on LoCoMo?" finds no answer, only a crossfire of mutually irreconcilable numbers: Zep ∼84% (later corrected to 58.44%, counter-claimed at 75.14%), Mem0 67.13% and later ∼92.5%, independent reports at 75.78% and 91.69%. Paper 1 [Gurbanov, 2026] argues — from primary sources — that this is **a measurement artifact, not an architecture verdict**, and that the spread resolves into four named forks: (A) denominator/adversarial-category handling, (B) who-configured-the-loser, (C) judge- and answer-model drift, and (D) benchmark-identity confusion. Paper 1 proposes one frozen protocol that neutralizes all four.

But Paper 1's load-bearing premise — *that protocol choices alone move the number by margins comparable to the architecture gaps vendors compete over* — is, in Paper 1, supported by forensics plus a single live pilot data point (a correct long-context answer that scored 0.201 under token-F1 — the real pilot longcontext token_f1 from `results/pilot_p01/raw.jsonl`). That is an existence proof, not a response surface. **This paper is the controlled experiment that turns that premise into a measured quantity.** We hold the memory system fixed and vary only the evaluation protocol, treating the protocol as a factorial design over six axes: **judge model, judge prompt, answer prompt, adversarial/denominator handling, retrieval-k, and seed.**

The problem in one sentence. Reported memory-benchmark scores are not comparable, because the number a paper prints is a function of protocol choices it usually does not report. Two recent voices name this without measuring it. Vectorize [2026] asserts that "small changes can swing accuracy scores by double digits" but supplies no ablation, no variance, no seeds, no surface. Penfield Labs [2026] audits LoCoMo's answer key (≈6.4% wrong) and shows one `gpt-4o-mini` judge accepts up to ≈63% of intentionally-wrong answers — a real finding about one judge configuration that does not vary the judge model or prompt and does not quantify the accuracy swing as a function of protocol. We fill exactly that gap.

Contributions. (1) The first *controlled factorial* sensitivity surface for memory-benchmark scoring on real LoCoMo. (2) A *rank-stability* analysis (Kendall's τ and pairwise-swap fraction) that asks whether swapping only the judge reorders the leaderboard — Stage 2 PENDING. (3) A *memory-eval reproducibility checklist* — the minimum set of protocol facts a paper must report for its number to be reproducible to within a stated tolerance — released as an adoptable artifact (§6).

Relationship to Paper 1. Same harness, same real LoCoMo data, different claim and different table. Paper 1 *adjudicates* a vendor dispute (forensics + one fixed protocol); Paper 2 *measures* the response surface that makes adjudication necessary. Neither scoops the other.

2 Background & Related Work *[DRAFTED]*

LoCoMo. Maharana et al. [2024] (ACL 2024; arXiv:2402.17753; `snap-research/locomo`, CC BY-NC 4.0, commit `3eb6f2c`). Ten multi-session conversations, ∼300 turns/∼9K tokens each, questions tagged single-hop, multi-hop, open-domain, temporal, and adversarial (Cat 5). The repository's official primary metric is SQuAD token-overlap F1; standard practice scores Categories 1–4 (N=1540 here). LoCoMo is the dataset whose protocol choices we vary.

Adjacent benchmarks. Wu et al. [2024] (LongMemEval, arXiv:2410.10813) and DMR are different datasets with different protocols. Paper 1 documents that quoting one as if it were a LoCoMo number is Fork D. We treat one benchmark per table; this draft's measured numbers are LoCoMo-only.

Memora/FAMA. Uddin et al. [2026] (arXiv:2604.20006). Adopted prior art for the project's metric work; a methodological neighbor for judge-style correctness scoring our Stage-2 judge axis will vary.

Assert-only predecessors (the gap we fill). Vectorize [2026]: asserts the swing; no measurement. Penfield Labs [2026]: audits the key + one judge; no cross-model/prompt sensitivity. We cite both as motivation; we reuse Penfield's "adversarial wrong answer" idea as one Stage-2 validation probe (PENDING).

LLM-as-judge sensitivity (general). Bellibatlu et al. [2026] (arXiv:2604.23478) and Prompt Sensitivity Study Authors [2025] (arXiv:2509.01790) establish the method of treating the judge as a variable; our novelty is **localizing it to the memory-benchmark setting** and tying it to the **system-ranking consequence inside a live vendor dispute**.

3 Method *[DRAFTED]*

3.1 The controlled factorial sweep

We fix the memory system and treat the evaluation protocol as a factorial over six axes, none of which is part of the system under test:

1. **Judge model** — which LLM scores answer-vs-gold. *(Stage 2, PENDING.)*
2. **Judge prompt** — strict/lenient/CoT-rubric wording. *(Stage 2, PENDING; `eval/prompts/judge_{strict,l`*
3. **Answer prompt** — concise vs. verbose generation instruction. *(Stage 2, PENDING; `eval/prompts/answer_{`*
4. **Adversarial/denominator handling** — Cat-5 inside or outside the denominator. *(Stage 1, FREE.)*
5. **Metric** — exact-match vs. token-F1 vs. normalized variants. *(Stage 1, FREE.)*
6. **Retrieval-k** — how many passages are surfaced, $k \in \{1, 3, 5, 10\}$. *(Stage 1, FREE.)*
7. **Seed** — judge/answer stochasticity. *(Stage 2, PENDING.)*

Stage 1 ($0, no LLM) covers axes 4–6: pure re-aggregation of existing scored outputs plus free lexical scorers over the free `bm25`/`tfidf_rag` retrieval outputs. **Stage 2 (paid, ∼$30–60, deferred)** covers axes 1–3 and 7.

The Stage-1 driver is `eval/sweep.py`: it wraps the existing free retrievers and `eval.metrics` **without editing** `runner.py`/`judge.py`/ `answer_gen.py`/`metrics.py`/any `systems/*.py` (a paid vendor pilot uses those files concurrently; this study is additive-only). It writes `results/sweep/stage1.json` and `results/sweep/swing.csv`; the manifest records `total_cost_usd`, `llm_calls`, and a `zero_cost_asserted` flag, and `run_sweep()` **raises** if cumulative spend is ever non-zero.

Reproduce Stage 1 ($0):

```
python -m eval.sweep --tasks data/external/locomo \
    --systems bm25,tfidf_rag --ks 1,3,5,10 --out results/sweep
python -m pytest -q tests/test_sweep.py   # 7 invariant tests; full suite 47 passed
```

3.2 The sensitivity metric (swing)

For an axis A with settings $\{a_1, \ldots, a_m\}$, holding all other axes fixed at a reference configuration, the **swing** of a fixed system s is:

$$\mathrm{swing}(s, A) \;=\; \max_i \mathrm{score}(s;\, a_i) \;-\; \min_i \mathrm{score}(s;\, a_i).$$

We report swing in **absolute points** and, where meaningful, **relative** terms (swing/reference score). A swing is *protocol-induced* by construction: the system s is identical across all a_i; only the measurement choice moves.

3.3 Rank stability *[STAGE-2 PENDING]*

For each pair of judge configurations (model × prompt), we will compute **Kendall's** τ between the two induced system rankings and the **fraction of pairwise swaps**; $\tau < 1$ means swapping only the judge reorders the leaderboard. Requires real judge calls; **Stage 2 (PENDING)**; no rank-stability number is reported in this draft.

3.4 Systems and data

Stage 1 runs the keyless extractive retrievers `bm25` and `tfidf_rag` over real `snap-research/locomo` @ `3eb6f2c`: **1540 answerable** QA (Cat 1–4: single_hop 841, temporal 321, multi_hop 282, open_domain 96) plus **446 adversarial** (Cat 5). The object of study is the *swing*, not the level. Stage 2 adds generation-based sources so that the judge axes move a real answer score.

4 Results §A — Stage-1 FREE Axes (real LoCoMo, $0, 0 LLM calls) *[STAGE-1 REAL]*

All numbers below are computed on real LoCoMo by re-aggregating existing scored outputs; total cost **$0.00**, **0** LLM/Anthropic calls (manifest: `total_cost_usd=0.0, llm_calls=0, zero_cost_asserted=true`
None of these axes touch the system under test.

4.1 Axis 6 — Retrieval-k (the largest measured swing)

Recall@k of the answer-bearing passage, $k \in \{1, 3, 5, 10\}$, system held fixed:

Table 1: Retrieval-k sensitivity: Recall@k, system fixed (LoCoMo Cat 1–4, $0.00).

System	k=1	k=3	k=5	k=10	**swing (range)**
bm25	0.2558	0.4071	0.4786	0.5597	**0.3039 (30.4 pts)**
tfidf_rag	0.2169	0.3753	0.4448	0.5390	**0.3221 (32.2 pts)**

A fixed retriever's "did it surface the evidence" score **more than doubles** (∼2.5×) purely by moving k from 1 to 10 — tf-idf moves **21.7% → 53.9%** and bm25 **25.6% → 56.0%** — *with no change whatsoever to the system.*

Honest sub-finding. For these extractive systems the answer *string* is always the top-1 passage, so the metric *score* on the answerable set is k-invariant (`retrieval_k_score` range = 0.0 across all k and all metrics in `swing.csv`). Retrieval-k moves *recall*, not the extractive answer score. Under the paid Stage-2 `generate` answer mode, where the answer model reads the top-k context, k will move the *answer* score too; that cell is **PENDING**.

4.2 Axis 5 — Metric (identical predictions, k fixed)

Varying only the scoring metric on the **identical** predictions:

Table 2: Metric sensitivity: identical predictions, only the scorer changes (LoCoMo Cat 1–4, \$0.00).

System	exact_match	token_f1	norm_contains	norm_em_first	**swing**
bm25	0.0000	0.0596	0.0727	0.0091	**0.0727 (7.3 pts)**
tfidf_rag	0.0000	0.0520	0.0643	0.0078	**0.0643 (6.4 pts)**

The metric choice swings the reported number by up to **7.3 absolute points** on unchanged answers, and the **relative swing is unbounded**: strict `exact_match` reports **0.0%** where `normalized_contains` reports **7.3%** — the metric alone decides whether a system "scores zero."

4.3 Axis 4 — Cat-5 Denominator Fork (the denominator component of Fork A, measured)

LoCoMo's answerable set is 1540 (Cat 1–4); 446 adversarial Cat-5 items use the official abstention metric. The extractive retrievers always emit a non-empty answer, so their Cat-5 abstention accuracy is 0.0000; pooling Cat-5 into the denominator multiplies the headline by $1540/1986 = 0.7754$:

Table 3: Cat-5 denominator fork: fixed system, fixed metric, only denominator scope changes.

System	metric	excl. Cat-5	incl. Cat-5	swing	rel. drop
bm25	token_f1	0.0596	0.0462	0.0134	−**22.5%**
bm25	norm_contains	0.0727	0.0564	0.0163	−**22.5%**
tfidf_rag	token_f1	0.0520	0.0403	0.0117	−**22.5%**
tfidf_rag	norm_contains	0.0643	0.0498	0.0144	−**22.5%**

The denominator fork is a deterministic −**22.46% relative haircut** on whatever the answerable-set score is — the **denominator component** of Paper 1's Fork A. On a published 84%-style number the same 0.7754 multiplier gives the $84 \to 65.1$-style move; the **residual to 58.44 is an unmeasured Stage-2 judge effect**, not this fork. Here we *measure* the multiplier directly and deterministically, on real data; the full fork-A reconstruction is not claimed.

4.4 Stage-1 headline (RQ1, free axes)

Holding the system fixed, the largest non-architecture, protocol-only swing is the **retrieval-k axis** (tf-idf **+0.3221**, bm25 **+0.3039**, ∼2.5×). The **metric axis** flips a system between **0.0% and 7.3%** on identical answers (unbounded relative swing). The **Cat-5 denominator fork** is a deterministic **−22.5% relative haircut**. None of these touch the system under test — and the single largest (32.2 points) is already of the same order as the architecture gaps vendors dispute, supporting Paper 1's premise even before the paid judge axis is run.

5 Results §B — Stage-2 PAID Axes (judge model/prompt; rank-flip) *[STAGE-2 REAL — pilot scope]*

`eval/rejudge.py` re-judged the **same** pre-generated answers from `results/pilot_p01/raw.jsonl` (3 systems × 40 Q × persona p01, pilot scope) under 5 judge configurations — **without re-generating any answer**. Sonnet-default verdicts reused from the pilot at \$0; 4 paid configs × 120 calls = 480 API calls, **total cost \$0.92** (hard budget cap: \$15). Outputs: `results/stage2_rejudge/`.

Scope caveat: Pilot p01 only — 3 systems, 40 Q, single persona. mem0/zep free-tier excluded (unreliable). Full N camera-ready upgrade.

Table 4: Judge sensitivity surface (LoCoMo pilot p01; J = judge-accuracy on 40 Q). [STAGE-2 REAL]. Sonnet/default is the reference config (pilot verdicts reused, $0).

Answer source	Sonnet/ default (ref)	Sonnet/ strict	Sonnet/ lenient	Sonnet/ CoT-rubric	Haiku/ default	**judge swing**
longcontext	0.775	0.625	0.850	0.575	0.650	**0.275 (27.5 pts)**
tfidf_rag	0.025	0.025	0.150	0.025	0.025	**0.125 (12.5 pts)**
bm25	0.025	0.025	0.150	0.025	0.025	**0.125 (12.5 pts)**
mem0 (free-tier)	**PENDING** — excluded (free-tier unreliable)					**PENDING**
zep (free-tier)	**PENDING** — excluded (free-tier unreliable)					**PENDING**

The **judge-prompt swing for longcontext = 27.5 absolute points** (57.5% under CoT-rubric vs. 85.0% under lenient) on identical, fixed answers. The lenient prompt raises bm25/tfidf_rag from 2.5% to 15.0% (+12.5 pts) by accepting partial/paraphrase answers the default rejects — the measurement-leniency confound that inflates vendor-reported numbers.

Table 5: Rank stability under judge-only changes (reference: Sonnet-default). [STAGE-2 REAL]. Reference ranking: longcontext > bm25 = tfidf_rag.

Judge-pair compared	Kendall's τ	pairwise-swap fraction	ranking flips?
Sonnet-default vs. Haiku-default	1.000	0.000	**No — stable**
Sonnet-default vs. Sonnet-strict	1.000	0.000	**No — stable**
Sonnet-default vs. Sonnet-lenient	1.000	0.000	**No — stable**
Sonnet-default vs. Sonnet-CoT-rubric	1.000	0.000	**No — stable**

The ranking longcontext > bm25 = tfidf_rag holds under all 4 judge perturbations ($\tau = 1.0$, 0 pairwise swaps). The gap between longcontext and the retrieval baselines is large enough ($\geq$37.5 pts at minimum) that no judge variant closes it in this pilot. *However*: with a 27-pt judge swing on longcontext and systems within 10–15 pts of each other, this same measurement variance would flip the ranking — the pilot's wide-gap geometry makes rank-stability a necessary but not sufficient condition.

Remaining Stage-2 tables (PENDING):

- Table B3: Answer-prompt × judge interaction (concise vs. verbose). **PENDING**
- Table B4: Penfield validation probe — fraction of intentionally-wrong answers accepted across the judge grid. **PENDING**
- Table B5: Seed variance — J mean ± std over $\geq$5 seeds per cell. **PENDING**

6 A Memory-Eval Reproducibility Checklist (the adoptable artifact) *[DRAFTED]*

The minimum set of protocol facts that must be reported for a memory-benchmark number to be reproducible. Each item is grounded in a measured (Stage 1) or to-be-measured (Stage 2) swing axis; the parenthetical is the swing it controls.

1. **Dataset + commit + answerable set** — exact corpus, version/commit, and N (e.g. `snap-research/locomo` @ `3eb6f2c`, Cat 1–4, N=1540). *(Pins the denominator base.)*
2. **Denominator/Cat-5 handling** — is the adversarial category in or out of the denominator? *(Axis 4 — deterministic ±22.5% relative move; Stage-1 measured.)*
3. **Metric** — exact-match vs. token-F1 vs. normalized variant, defined exactly. *(Axis 5 — up to 7.3 absolute points and unbounded relative swing on identical answers; Stage-1 measured.)*
4. **Retrieval-k** — how many passages are surfaced. *(Axis 6 — up to 32.2-point ∼2.5× recall swing; Stage-1 measured.)*
5. **Answer model + answer prompt** — the generation model and its instruction. *(Axis 3 — Stage-2 PENDING.)*
6. **Judge model + judge prompt** — the scoring model and its rubric. *(Axes 1–2 — Stage-2 REAL (pilot): judge swing 27.5 pts for longcontext, 12.5 pts for baselines; ranking stable τ=1.0 across all 4 perturbations in the pilot.)*
7. **Seeds + dispersion** — number of seeds and mean ± std. *(Axis 7 — Stage-2 PENDING.)*
8. **Manifest** — released config, seeds, git SHA, dataset commit, and per-cell results so any single number is independently re-runnable.

A paper reporting all eight makes its memory-benchmark number reproducible to within a stated tolerance; omitting any one leaves a swing axis unpinned. The checklist plus the released sweep configs and tidy results CSV (`results/sweep/`) are the reusable artifact.

7 Limitations / Threats to Validity *[DRAFTED]*

- **Extractive Stage-1 systems → low absolute base.** The free retrievers are a retrieval floor; the object of study is the *swing*, not the level. The retrieval-k axis moves *recall*, not the extractive answer score.
- **Stage-2 pilot scope.** The judge-sensitivity results (Tables B1–B2) are from the p01 pilot: 3 systems, 40 Q, single persona. The rank-stability verdict (τ=1.0, stable) holds for this wide-gap configuration (longcontext leads by ≥37.5 pts); it does *not* generalize to configurations where systems are within 10–15 pts, where a 27-pt judge swing would flip the ranking. Full N camera-ready upgrade.
- **Single judge family (Stage 2).** The judge models run are `claude-sonnet-4-5` (4 prompt variants) and `claude-haiku-4-5`; a cross-provider judge (`gpt-4o-mini`, bridging to the historical vendor regime) is named as the intended check, not run (Anthropic-only key).
- **Pilot N=40 per system.** Confidence intervals are wide (±5–7 pts); the qualitative pattern (27-pt judge swing on a generation system, stable ranking for wide-gap configs) is the finding; exact J values carry ±5–7 pt uncertainty.
- **Judge-on-judge circularity.** The rank-stability analysis is designed to *expose* the confound ($\tau < 1$), not to escape it.
- **LoCoMo-only measured numbers.** LongMemEval is named as the extension axis; one benchmark per table.

- **`git_sha = "unknown"` in the manifest.** This worktree has no resolvable HEAD commit, matching the existing harness's own recorded value — the honest value, not a sweep bug.

8 Reproducibility Statement *[DRAFTED]*

Stage 1 reproduces today at **$0.00**: `eval/sweep.py` imports no Anthropic SDK and makes no model calls; every retriever query reports `cost_usd == 0.0`; `run_sweep()` raises if cumulative spend is ever non-zero; `results/sweep/stage1.json` records `total_cost_usd=0.0, llm_calls=0, zero_cost_asserted` The driver is additive-only. Tests: `python -m pytest -q` → **47 passed** (40 pre-existing + 7 new in `tests/test_sweep.py`).
Stage 2 (`eval/rejudge.py`, additive-only — does not edit the above files) ran at **$0.92**: 480 paid judge calls (4 configs × 120 answers), Sonnet-default reused at $0. All inputs taken from `results/pilot_p01/raw.jsonl` (pre-generated answers); no new answer generation. Hard budget cap $15. Released artifacts: `results/stage2_rejudge/{cells.jsonl,summary.json}`. Answer-prompt and seed axes remain **PENDING**.

9 Venue Plan *[DRAFTED]*

Now — arXiv, bundled with Paper 1. Release the audit (forensics + protocol) and this sensitivity surface together so each cites the other and the pair reads as one coherent program: *memory-eval is broken — here is the forensics AND the controlled measurement.*

After Stage 2 (judge-swap + rank-flip + checklist validated) — an LLM-evaluation workshop at EMNLP 2026 or NeurIPS 2026, with a Findings/short-paper track at ACL/EMNLP 2026 as the stretch if the rank-flip is strong and N is scaled. Workshop + arXiv is the bankable plan; main-track is a stretch solo.

10 Conclusion

Reported memory-benchmark scores are not comparable because the number a paper prints is a function of protocol choices it usually does not report. We turn this assertion into a measurement: holding the system fixed and varying only the evaluation protocol, the largest Stage-1 (free) swing is **32.2 absolute points** ($\sim$2.5×) from the retrieval-k axis alone; the metric axis produces an unbounded relative swing (0.0% vs. 7.3% on identical answers); and the Cat-5 denominator fork is a deterministic −22.5% relative haircut — the **denominator component** of the 84 → 58.44 fork: ×0.7754 alone takes 84 → 65.1; the residual to 58.44 is an unmeasured judge effect, not this fork. Stage-2 (pilot, 3 systems × 40 Q, $0.92) adds the judge axis: the **judge-prompt swing for longcontext is 27.5 absolute points** (CoT-rubric: 57.5% → lenient: 85.0%) on identical, fixed answers — the same order of magnitude as the architecture gaps vendors dispute. The **ranking is stable** ($\tau = 1.0$) in this wide-gap pilot configuration (longcontext leads by ≥37.5 pts); answer-prompt and seed axes remain **PENDING**. The memory-eval reproducibility checklist (§6) is released now as an adoptable artifact.

References

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